



PHISHING EMAIL DETECTION & AWARENESS SYSTEM

Cybersecurity Internship Project

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Phishing Email Detection and Awareness System

Introduction

Phishing is one of the most widespread and effective forms of cyberattacks used by cybercriminals to steal sensitive information such as usernames, passwords, banking details, and personal data.

These attacks are typically carried out through fraudulent emails that imitate trusted organizations, including banks, online service providers, government agencies, or company IT departments.

By creating a false sense of urgency or trust, attackers attempt to persuade recipients to click malicious links, download infected attachments, or disclose confidential information.

Objective

The goal of this project is to analyze phishing email samples, identify common phishing indicators, classify each email as Safe, Suspicious, or Phishing, explain the techniques used by attackers, and provide practical recommendations that help users recognize and prevent phishing attacks.

- Analyze phishing email samples to identify suspicious characteristics.
- Identify common phishing indicators, such as spoofed senders, fake domains, malicious links, and suspicious attachments.
- Classify emails into **Safe**, **Suspicious**, or **Phishing** based on the analysis.
- Explain how phishing attacks work and the techniques attackers use to deceive users.
- Provide practical prevention measures and security awareness guidelines to help users recognize and avoid phishing attempts.

Tools Used

The following tools and resources were used to analyze phishing emails, identify potential threats, and prepare the project documentation:

- **Web Browser (Google Chrome / Microsoft Edge):** Used to inspect URLs, verify domains, and access online security analysis tools.
- **Email Header Analyzer:** Used to examine email headers, verify sender information, and analyze email authentication mechanisms such as SPF, DKIM, and DMARC.
- **Documentation Tools (Google Docs / Microsoft Word):** Used to prepare and format the project report before exporting it as a PDF.

- **Public Phishing Email Samples and Datasets:** Used to study real-world phishing emails and compare them with legitimate emails for analysis and classification.

Example email header analysis tools used for investigation:

- <https://toolbox.googleapps.com/apps/messageheader/>

Phishing Email Sample Analysis

Email Example 1 – Promotional Banking Email (Jupiter Salary Account Upgrade)

From: Jupiter noreply@jupiter.money

Date: Mon, 6 Jul 2026 at 16:25

Subject: Upgrade to Jupiter Salary Account - Get Guaranteed Rewards

Identified Indicators

Indicator	Explanation
Sender Email	The sender domain matches the official Jupiter domain, indicating consistency with the claimed organization.
Subject Line	The subject promotes a salary account upgrade with rewards. Promotional language alone is not a phishing indicator.
SPF Authentication	The sender's mail server is authorized to send emails on behalf of the domain, increasing confidence in the email's authenticity.
DKIM Authentication	The email's digital signature is valid, indicating that the message content was not altered during transmission.
DMARC Authentication	Although DMARC failed, this can occur because of configuration issues, forwarding, or alignment problems. A DMARC failure alone does not confirm phishing when SPF and DKIM both pass.

Risk Classification

- Classification: Suspicious (Low Risk)

Reason:

The email successfully passes both SPF and DKIM authentication, which are strong indicators of authenticity. However, the DMARC failure introduces a minor concern that should be investigated further. Without additional phishing indicators such as spoofed domains, malicious links, urgent requests, or credential theft attempts, the email cannot be classified as phishing based solely on the available information.

Attack Explanation

No clear phishing attack is identified from the available evidence. If this email were malicious, an attacker might attempt to use promotional offers to encourage users to click links or provide personal information. Therefore, users should always verify that links point to the official domain before clicking and avoid sharing sensitive information unless they have confirmed the legitimacy of the email.

Email Example 2 – Job Recruitment Email Analysis

From: Ritu ritu@session.in

Date: Sat, 20 Jun 2026 at 10:42

Subject: Immediate Job Vacancies at ACCENTURE | Be Selected For Upcoming Interview Drive
Hundred Thirty Two

Identified Indicators

Indicator	Explanation
Sender Email	The email claims to offer jobs at Accenture , but the sender's domain is session.in , not an official Accenture domain. This mismatch should be verified.
SPF	The sending server is authorized to send emails for the session.in domain.

DKIM	The email content has not been modified during transit.
DMARC	The domain failed DMARC alignment. This increases suspicion but does not alone prove the email is malicious.
Organization Mentioned	If the sender is not officially associated with Accenture, the email should be treated cautiously. Users should verify job offers through official career portals.

Risk Classification

- **Classification: Suspicious (Medium Risk)**

Reason:

The email passes SPF and DKIM authentication. However, the sender domain (session.in) does not obviously match the organization mentioned in the subject (Accenture). The DMARC check failed. The urgent recruitment message could be a legitimate recruitment campaign or a social engineering attempt. Additional verification of the sender, links, and email content is recommended before interacting with the email.

Attack Explanation

This email demonstrates a **potential social engineering technique** commonly used in recruitment-related phishing campaigns. Attackers may impersonate well-known companies or advertise attractive job opportunities to persuade users to click links, submit resumes, or disclose personal information. Although this email is **not conclusively phishing** based on the header analysis alone, recipients should verify the legitimacy of the recruitment offer through the company's official careers website before responding.

Email Example 3 – Interview Invitation

From: Kanikha HR info@skill.getujobs.com

Date: Sat, 16 May 2026 at 20:17

Subject: TECH MAHINDRA (Video Call) INTERVIEW Archana S | Your Application has been shortlisted

Identified Indicators

Indicator	Explanation
Sender Domain	The email claims to relate to Tech Mahindra , but it is sent from a different domain. This may be a recruitment partner, but it should be verified.
Subject Line	Uses attractive recruitment language to encourage the recipient to open the email.
Brand Mismatch	Users should verify whether the sender is officially associated with Tech Mahindra before responding.
SPF Authentication	The sending server is authorized for the sender's domain.

Risk Classification

- Classification: Suspicious (Medium Risk)

Reason

Although the email passes **SPF**, **DKIM**, and **DMARC** authentication, the sender domain (skill.getujobs.com) does not directly belong to **Tech Mahindra**. Recruitment agencies and job portals often send interview notifications on behalf of employers, so this does not automatically indicate phishing. However, recipients should verify the legitimacy of the interview invitation through the official Tech Mahindra careers website or by contacting the recruiter before sharing personal information.

Attack Explanation

This email demonstrates a **potential recruitment-based social engineering scenario**. Cybercriminals frequently exploit well-known company names to attract job seekers and persuade them to click links, download attachments, or submit personal information. Even when authentication checks pass, users should verify the sender's relationship with the organization mentioned in the email before taking any action.

Images:-

Email 1 :- Promotional Banking Email

MessageId	1361928347.198933.1783335308554@ip-10-12-48-75.ap-south-1.compute.internal
Created at:	7/6/2026, 4:25:15 PM GMT+5:30 (Delivered after)
From:	Jupiter <noreply@jupiter.money> Using NetcoreCloud Mailer
To:	archanasp384@gmail.com
Subject:	Upgrade to Jupiter Salary Account - Get Guaranteed Rewards
DKIM:	pass with domain jupiter.money Learn more

#	Delay	From *	To *	Protocol	Time received
0		FMTA1-163.ncdelivery04.com.	→ [Google] mx.google.com	ESMTPS	7/6/2026, 4:25:15 PM GMT+5:30
1			→ [Google] 2002:a05:620a:1d07:b0:92e:4a63:d7ef	SMTP	7/6/2026, 4:25:15 PM GMT+5:30

Email 2 :- Job Recruitment Email Analysis

MessageId	302510419520394546@ncdelivery.session.in
Created at:	6/20/2026, 10:42:38 AM GMT+5:30 (Delivered after 1 sec)
From:	Ritu <ritu@session.in> Using NetcoreCloud Mailer
To:	archanasp384@gmail.com
Subject:	archanasp384 - Immediate Job Vacancies at ACCENTURE Be Selected For Upcoming Interview Drive HundredThirtyTwo
DKIM:	pass with domain session.in Learn more

#	Delay	From *	To *	Protocol	Time received
0		newmta6-40.143.etransmail.com.	→ [Google] mx.google.com	ESMTPS	6/20/2026, 10:42:38 AM GMT+5:30
1			→ [Google] 2002:a05:620a:4606:b0:914:bf8a:ba81	SMTP	6/20/2026, 10:42:38 AM GMT+5:30

Email Example 3 :- Interview Invitation

Google Admin Toolbox

Messageheader

Help

MessageId

1778942824-MjM5MjQ5NjUzMS0xNTI2NDkx@skill.getujobs.com

Created at:

5/16/2026, 8:17:04 PM GMT+5:30 (Delivered after 25 sec)

From:

"Kanikha HR" <info@skill.getujobs.com>

To:

archanasp384@gmail.com

Subject:

TECH MAHINDRA (Video Call) INTERVIEW Archana S | Your Application has been shortlisted

DKIM:

pass with domain skill.getujobs.com
[Learn more](#)

#	Delay	From *	To *	Protocol	Time received
0	24 sec	mail.mc1202.mlrcd.com.	→ [Google] mx.google.com	ESMTPS	5/16/2026, 8:17:28 PM GMT+5:30
1			→ [Google] 2002:a05:622a:1a93:b0:510:1325:32a0	SMTP	5/16/2026, 8:17:28 PM GMT+5:30

Risk Classification Summary

Email / Scenario	Category	Risk Level	Reason
Email 1 - Jupiter – Salary Account Upgrade (Promotional Banking Email)	Safe	Low	Sender domain matches the official organization. SPF and DKIM passed. Although DMARC failed, no additional phishing indicators were identified.
Email 2 - Accenture Recruitment	Suspicious	Medium	DMARC failed, and the sender domain does not directly match the organization mentioned in the email.
Email 3 - Tech Mahindra Interview	Suspicious	Medium	Brand mentioned differs from the sender's domain. Verification is recommended before responding.

Indicators Identified

During the analysis of the collected email samples, several common phishing indicators were identified. These indicators help distinguish legitimate emails from suspicious or malicious ones and play an important role in identifying potential phishing attempts.

- Fake or Suspicious Sender Domains
- Urgent or Threatening Language
- Suspicious LinksGeneric Greetings
- Requests for Sensitive Information
- Unexpected Attachments
- Brand and Domain Mismatch

Prevention Guidelines

- Users can reduce the risk of phishing attacks by following basic cybersecurity practices.
- Always verify the sender's email address and inspect website URLs before clicking links or entering sensitive information.
- Avoid opening suspicious links or downloading attachments from unknown sources, and never share passwords, OTPs, or confidential information through email.
- Enable Multi-Factor Authentication (MFA) to enhance account security, keep your devices and software up to date, and report suspicious emails to your organization's IT or security team.
- If an email appears unusual or unexpected, verify its authenticity through the organization's official website or customer support before taking any action.

Conclusion

Phishing attacks continue to be one of the most common cybersecurity threats, targeting users through deceptive emails and social engineering techniques. This project analyzed multiple email samples, identified key phishing indicators, and classified emails as **Safe**, **Suspicious**, or **Phishing** based on their characteristics. The analysis highlights the importance of verifying email authenticity, recognizing warning signs, and following cybersecurity best practices. Increasing user awareness and adopting preventive measures can significantly reduce the risk of phishing attacks and help protect both personal and organizational information